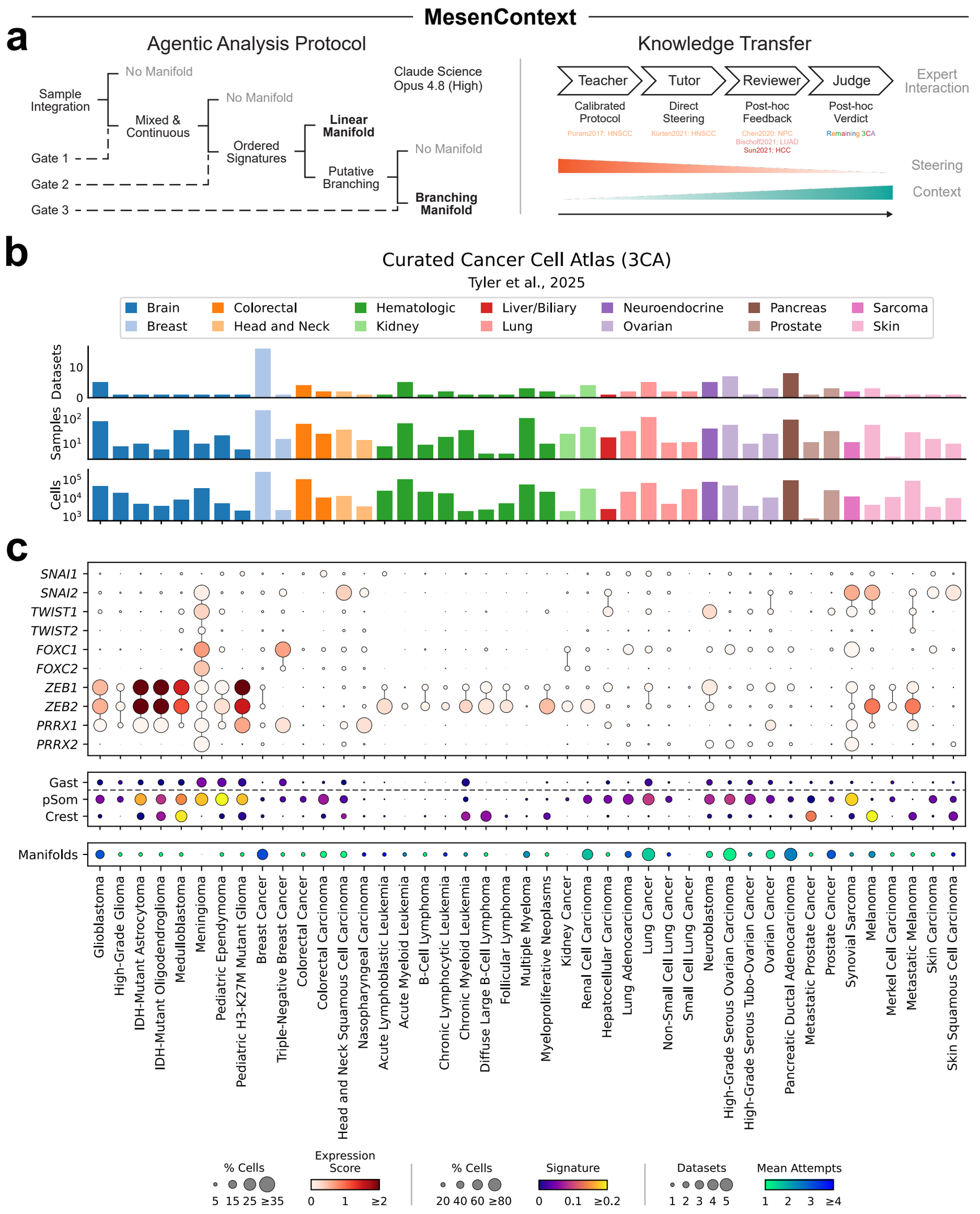

MesenContext
a
Agentic Analysis Protocol
Sample Integration
No Manifold
Mixed & Continuous
No Manifold
Ordered Signatures
Linear Manifold
Putative Branching
No Manifold
Branching Manifold
Gate 1
Gate 2
Gate 3
Claude Science Opus 4.8 (High)
Knowledge Transfer
Teacher
Tutor
Reviewer
Judge
Expert Interaction
Calibrated Protocol
Direct Steering
Post-hoc Feedback
Post-hoc Verdict
Puram2017: HNSCC
Kürten2021: HNSCC
Chen2020: NPC
Bischoff2021: LUAD
Sun2021: HCC
Remaining 3CA
Steering
Context
b
Curated Cancer Cell Atlas (3CA)
Tyler et al., 2025
Brain
Breast
Colorectal
Head and Neck
Hematologic
Kidney
Liver/Biliary
Lung
Neuroendocrine
Ovarian
Pancreas
Prostate
Sarcoma
Skin
Datasets
Samples
Cells
c
SNAI1
SNAI2
TWIST1
TWIST2
FOXC1
FOXC2
ZEB1
ZEB2
PRRX1
PRRX2
Gast
pSom
Crest
Manifolds
Glioblastoma
High-Grade Glioma
IDH-Mutant Astrocytoma
IDH-Mutant Oligodendroglioma
Medulloblastoma
Meningioma
Pediatric Ependymoma
Pediatric H3-K27M Mutant Glioma
Breast Cancer
Triple-Negative Breast Cancer
Colorectal Cancer
Colorectal Carcinoma
Head and Neck Squamous Cell Carcinoma
Nasopharyngeal Carcinoma
Acute Lymphoblastic Leukemia
Acute Myeloid Leukemia
B-Cell Lymphoma
Chronic Lymphocytic Leukemia
Chronic Myeloid Leukemia
Diffuse Large B-Cell Lymphoma
Follicular Lymphoma
Multiple Myeloma
Myeloproliferative Neoplasms
Kidney Cancer
Renal Cell Carcinoma
Hepatocellular Carcinoma
Lung Adenocarcinoma
Lung Cancer
Non-Small Cell Lung Cancer
Small Cell Lung Cancer
Neuroblastoma
High-Grade Serous Ovarian Carcinoma
High-Grade Serous Tubo-Ovarian Cancer
Ovarian Cancer
Pancreatic Ductal Adenocarcinoma
Metastatic Prostate Cancer
Prostate Cancer
Synovial Sarcoma
Melanoma
Merkel Cell Carcinoma
Metastatic Melanoma
Skin Carcinoma
Skin Squamous Cell Carcinoma
% Cells
5 15 25 ≥35
Expression Score
0 1 ≥2
% Cells
20 40 60 ≥80
Signature
0 0.1 ≥0.2
Datasets
1 2 3 4 5
Mean Attempts
1 2 3 ≥4